weight normalization

26/02/2026

1. Overview

Samples Used

Sample	Description	Source
Data	2018 UL SingleMuon	Real collision data
DY	Drell-Yan Z/gamma* to tau-tau	MC simulation
ttbar	tt to tau-tau + X	MC simulation
QCD	Multi-jet background	Data-driven (same-sign)

2. Processing Pipeline

```
NanoAOD --> Phase0 --> Phase1 --> Merge --> Proton Mixing --> Plots
```

Scripts

Stage	Data	DY	ttbar	QCD
Phase0	fase0_data_mutau.py	fase0_dy.py	fase0.py	-
Phase1	fase1_data.py	fase1_dy.py	fase1_ttjets.py	fase1_qcd.py

3. Phase0: initial skimming

Scripts

- `fase0_data_mutau.py` (Data)
- `fase0_dy.py` (Drell-Yan)
- `fase0.py` (ttbar)

Structure of scripts

1. **Trigger selection**: `HLT_IsoMu24` (single muon trigger)
2. **Luminosity filtering**: Only certified good runs (dadosluminosidade.txt)
3. **Object selection**: At least 1 muon and 1 tau
4. **Variable extraction**: Muon, Tau, Jets, MET kinematics

4. Phase1: Selection cuts

Variable	Cut	Description
muon_id	> 3	Medium MVA ID
tau_id1 (VSjet)	> 63	VeryTight DeepTau vs jets
tau_id2 (VSe)	> 7	Tight DeepTau vs electrons
tau_id3 (VSmu)	> 1	Loose DeepTau vs muons
muon_pt	> 35 GeV	muon transverse momentum
tau_pt	> 100 GeV	tau transverse momentum
charge product	< 0	opposite sign
Delta R	> 0.4	Angular separation

Note: QCD uses **same-sign** (SS) selection

5. Event weights in Phase1

Data

```
event_weight = 1.0  # Real data no correction needed
```

Drell-Yan (DY)

```
event_weight = generator_weight x mu_trig_sf x mu_idiso_sf x mu_reco_sf
```

ttbar

```
event_weight = 0.15 x mu_trig_sf x mu_idiso_sf x mu_reco_sf
```

Important: 0.15 already included - should be fixed later for consistency

QCD

```
event_weight = 1.0  # data driven
```

6. Where event weights are applied

Data (no proton mixing needed)

```
Phase0: weight=1.0 -> Phase1: weight=1.0 -> Plotting: 1.0
```

Drell-Yan

```
Phase1: event_weight = generator_weight x muon_SFs
Proton Mixing: weight = event_weight x 0.13
Plotting: final_weight = weight x 1.81
```

7. Where event weights are applied (cont)

ttbar

```
Phase1: event_weight = 0.15 x muon_SFs  <- 0.15 included here
Proton Mixing: weight = event_weight x 0.13
Plotting: final_weight = weight x 1.0
```

QCD (data-driven)

```
Phase1: weight=1.0 -> Plotting: final_weight=1.0
```

8. Cross-Section normalization factors

Formula used

```
weight = N_MC / N_expected
where N_expected = Luminosity x Cross-section
```

Values used

Sample	Factor	Included Where?
DY	1.81	Applied in plotting
ttbar	0.15	Already in Phase 1 event_weight

This does not change the outcome but should be fixed for consistency

9. Files

Processing scripts

Script	Purpose
fase0_data_mutau.py	Phase0 for Data
fase0_dy.py	Phase0 for DY
fase1_data.py	Phase1 for Data
fase1_dy.py	Phase1 for DY
fase1_ttjets.py	Phase1 for ttbar
merge_pp_mutau.py	Proton mixing
plot_m.py	Plotting

10. Proton acceptance measurement code

Method in ROOT:

```
TFile* f = TFile::Open("Data_merged_proton_vars.root");
TTree* tree = (TTree*)f->Get("tree");
Long64_t total = tree->GetEntries();
Long64_t good = tree->GetEntries(
    "Sum$(proton_multi_arm==0)>=1 && Sum$(proton_multi_arm==1)>=1"
);
double P = (double)good / (double)total;
cout << "Proton acceptance P = " << P << endl;
```

Result:

```
total events: 6,970
good events: 1,708
P = 0.245 (24.5%)
```

11. Single Lepton Plots: Cuts Applied

Proton requirements

Sample	Cut
Data/QCD	`Sum$(proton_multi_arm==0)>0 && Sum$(proton_multi_arm==1)>0`
MC (DY, ttbar)	`xi_arm1_1 >= 0 && xi_arm2_1 >= 0`
Signal	`xi_arm1_1 >= 0 && xi_arm2_1 >= 0`

12. Weights applied

Weights applied

Sample	Weight
Data	None
QCD	None
DY	Proton acceptance (0.245) - xsec normalization already included
ttbar	Proton acceptance (0.245) x Scale(0.15)
Signal	`weight` branch (lumi x xsec x SFs x rad. damage) x 5000 (visualization)

13. Weight fix: old vs new approach

Old weights (`MuTauPlots_weight_comparison.C`)

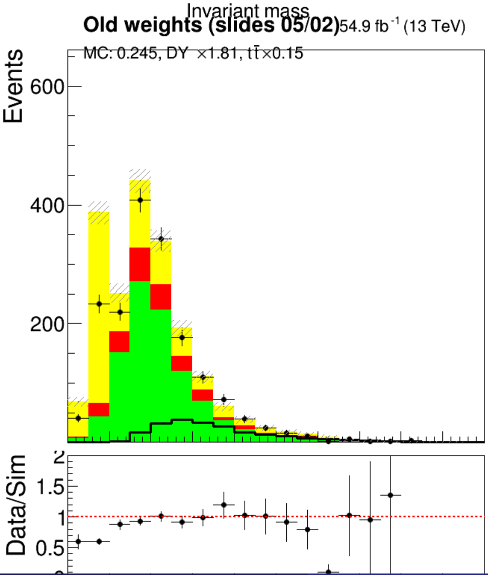
- MC filled with **flat** weight = `0.245` (proton acceptance, no `weight` branch)
- Then `Scale(1.81)` for DY, `Scale(0.15)` for ttbar
- No generator_weight, no muon scale factors

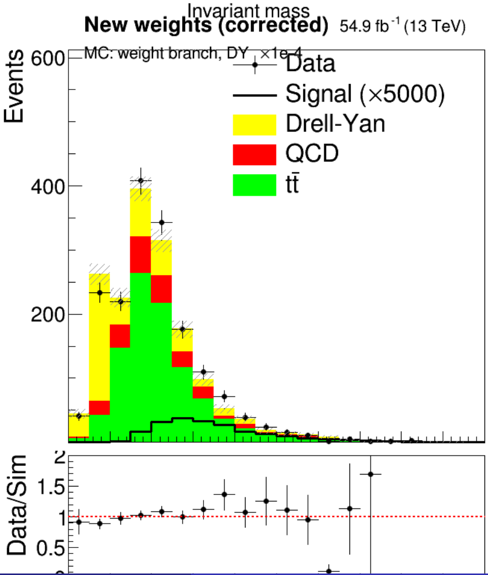
New weights (corrected)

- MC filled using **`weight` branch** = `generator_weight x muon_SFs x 0.245`
- Then `Scale(1.004e-4)` for DY (= L x sigma / Sum_w), `Scale(1.0)` for ttbar

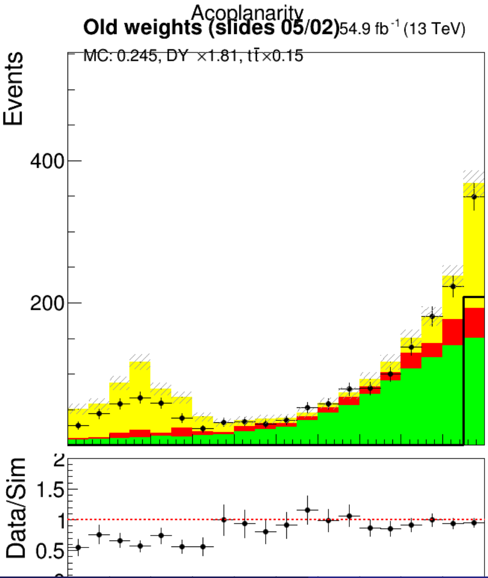
	Old	New
DY weight	`0.245 x Scale(1.81)`	`weight branch x Scale(1.004e-4)`
ttbar weight	`0.245 x Scale(0.15)`	`weight branch x Scale(1.0)`
DY Scale	1.81 (empirical)	1.004e-4 (= 54900 x 6077.22 / 3.323e12)
ttbar Scale	0.15 (empirical)	1.0 (already ~correct via 0.15 in Phase1)

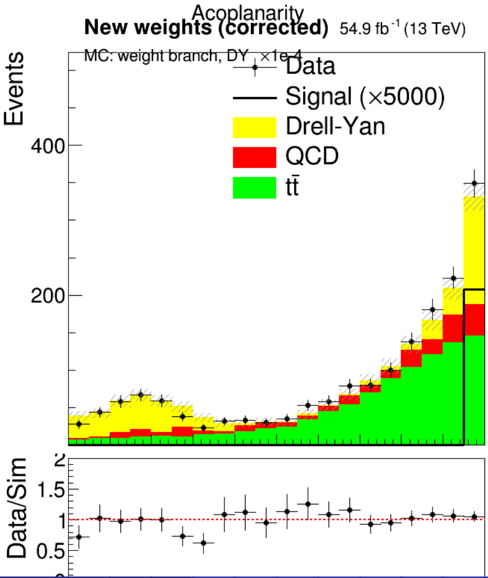
14. Weight comparison: Invariant Mass



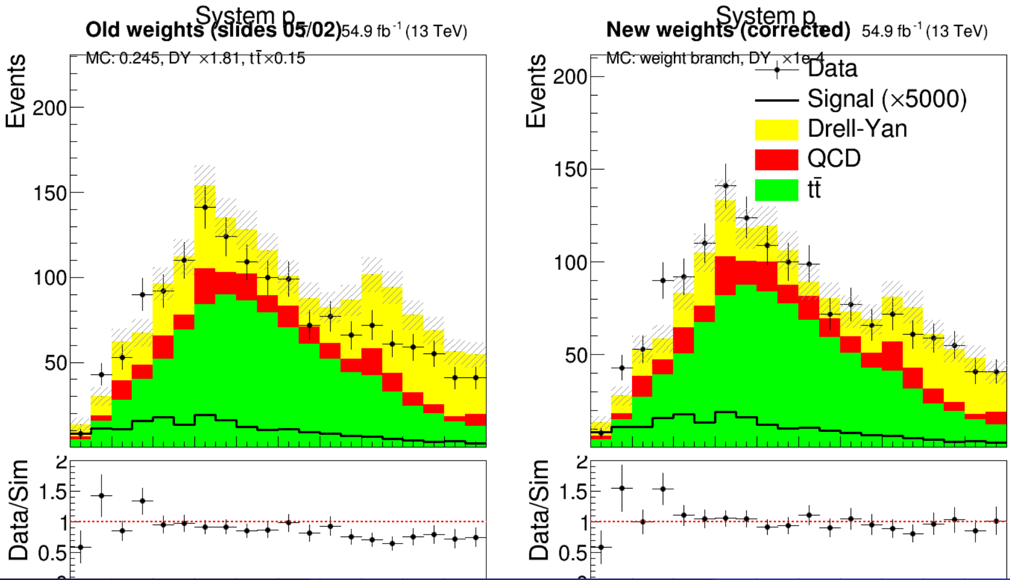


15. Weight comparison: Acoplanarity

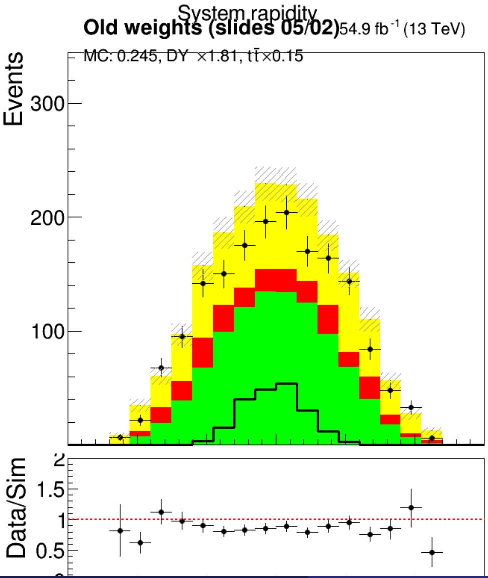


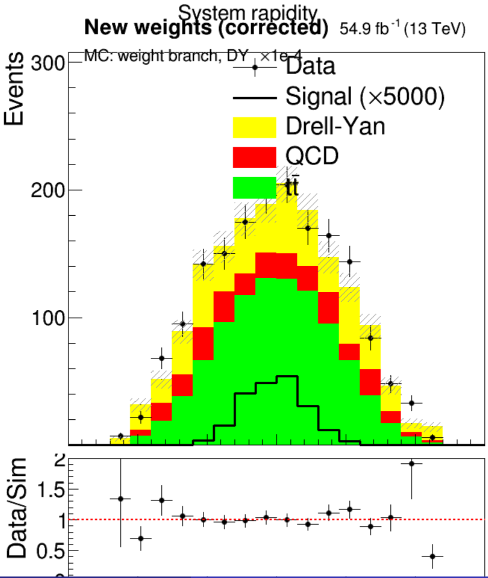


16. Weight comparison: System pT

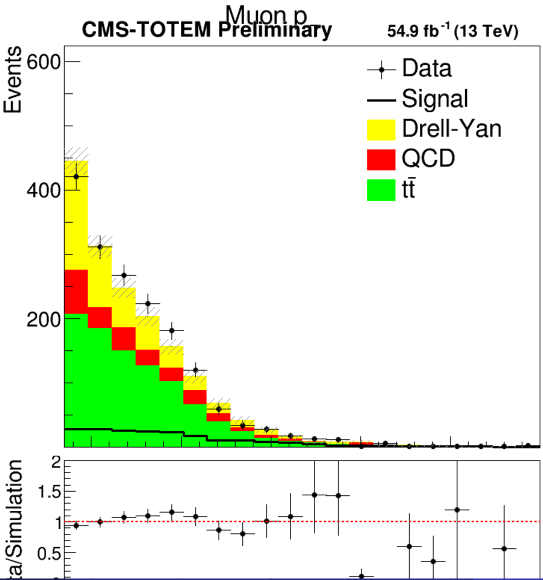


17. Weight comparison: System Rapidity

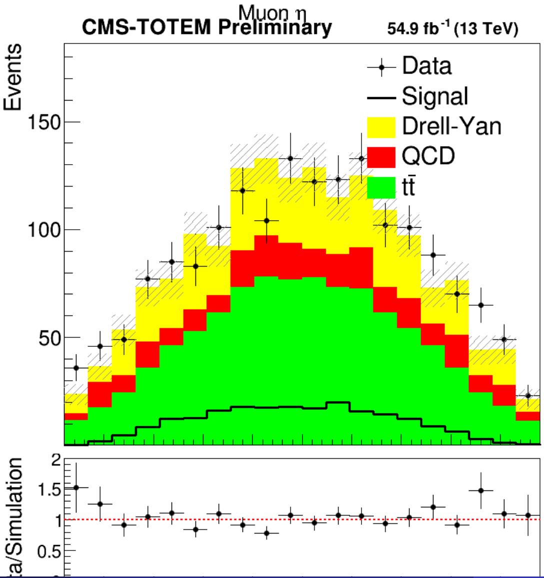




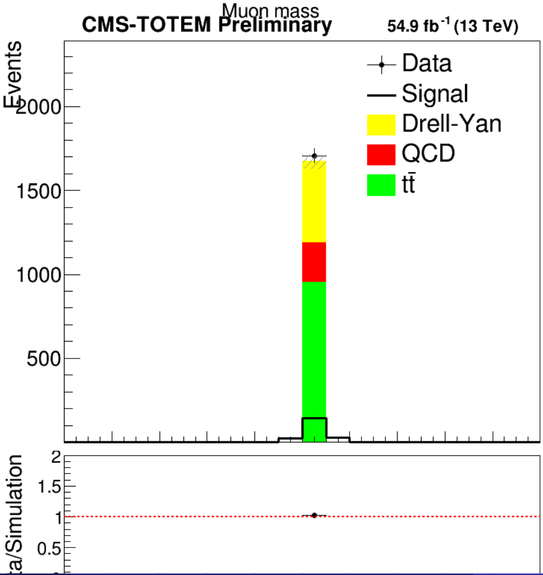
18. Muon pT



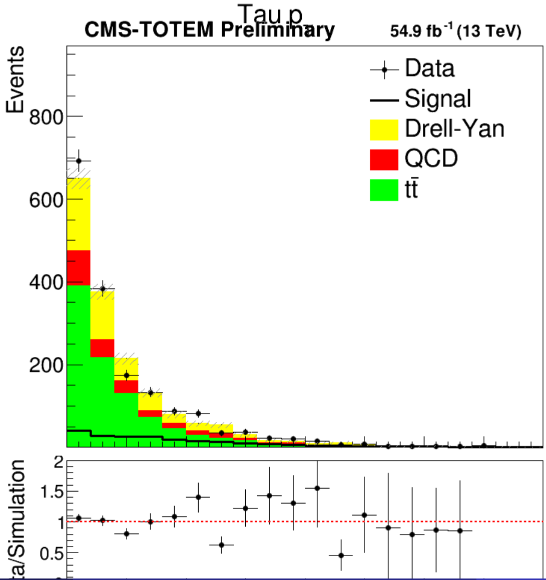
Muon p
CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
600
400
200
Data
Signal
Drell-Yan
QCD
$t\bar{t}$
ta/Simulation
2
1.5
1
0.5


Muon η
CMS-TOTEM Preliminary
54.9 fb⁻¹ (13 TeV)
Events
150
100
50
Data
Signal
Drell-Yan
QCD
tt̄
2
1.5
1
0.5
ta/Simulation

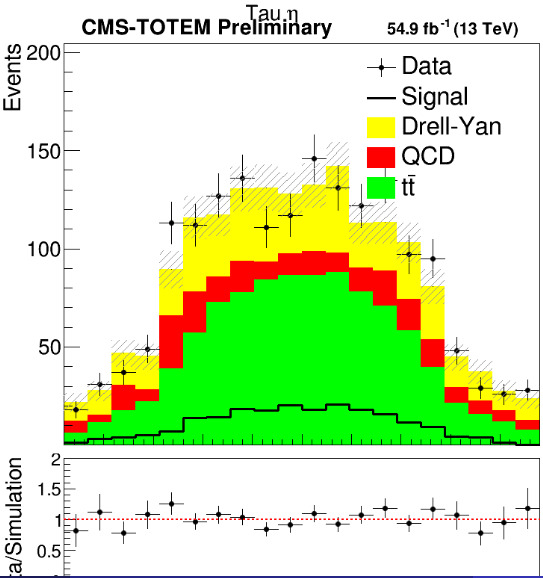
20. Muon mass


Muon mass
CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
2000
1500
1000
500
Data
Signal
Drell-Yan
QCD
$t\bar{t}$
2
1.5
1
0.5
ta/Simulation

21. Tau pT

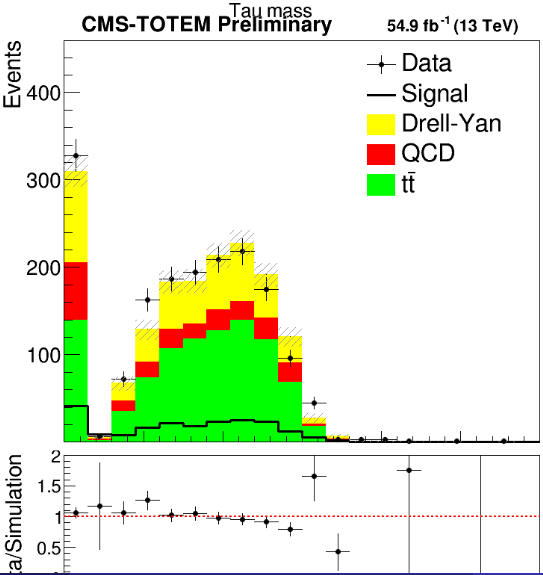

Tau p
CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
800
600
400
200
Data
Signal
Drell-Yan
QCD
$t\bar{t}$
2
1.5
1
0.5
ta/Simulation

22. Tau eta



Tau η
CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
200
150
100
50
Data
Signal
Drell-Yan
QCD
$t\bar{t}$
2
1.5
1
0.5
ta/Simulation

23. Tau mass


Tau mass
CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
400
300
200
100
Data
Signal
Drell-Yan
QCD
$t\bar{t}$
2
1.5
1
0.5

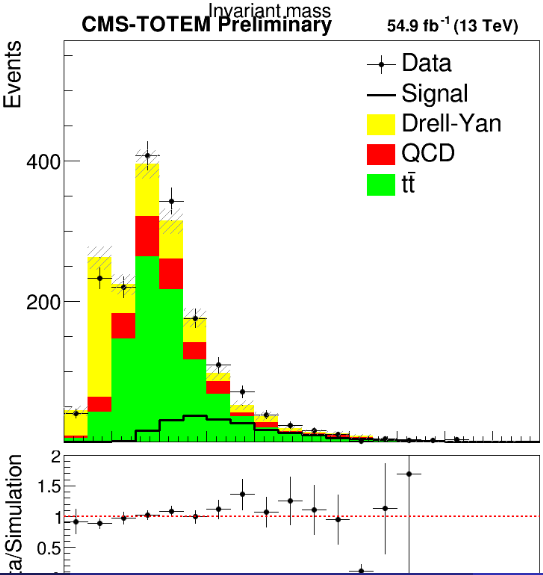
24. Single Distributions After Proton Merging

The following plots show individual kinematic distributions after requiring protons on both PPS arms.

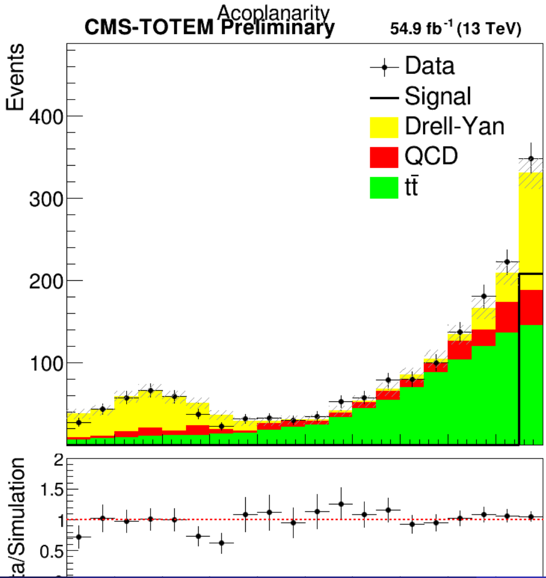
Variables shown

Variable	Description
`sist_mass`	Invariant mass m(mu,tau)
`acop`	Acoplanarity
`sist_pt`	System transverse momentum
`sist_rap`	System rapidity
`tau_pt`	Tau transverse momentum
`met_pt`	Missing transverse energy

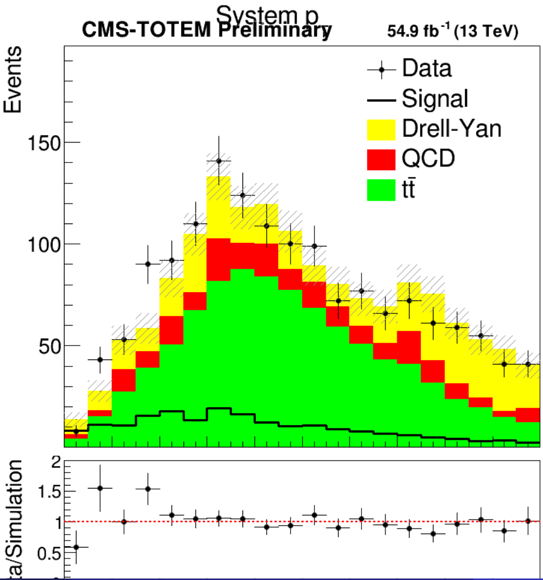
25. Invariant Mass (after proton requirements)



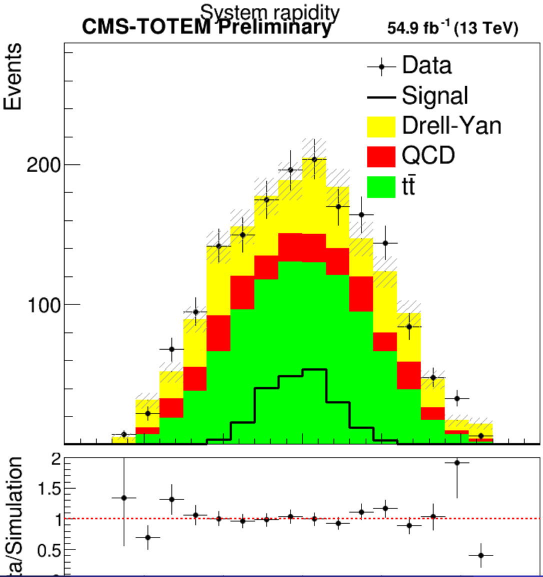
26. Acoplanarity (after proton requirements)



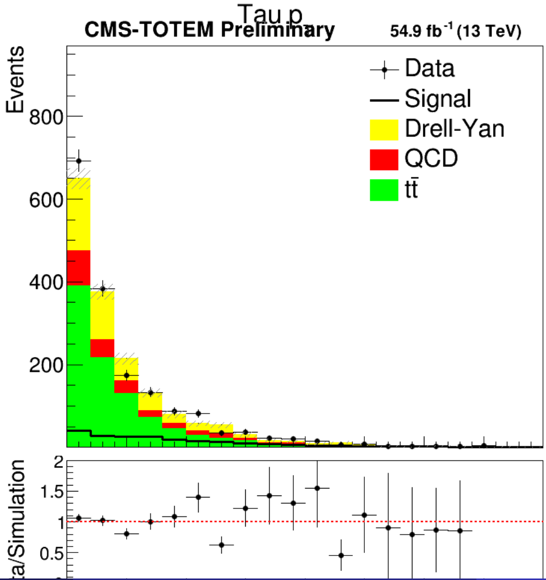
27. System pT (after proton requirements)



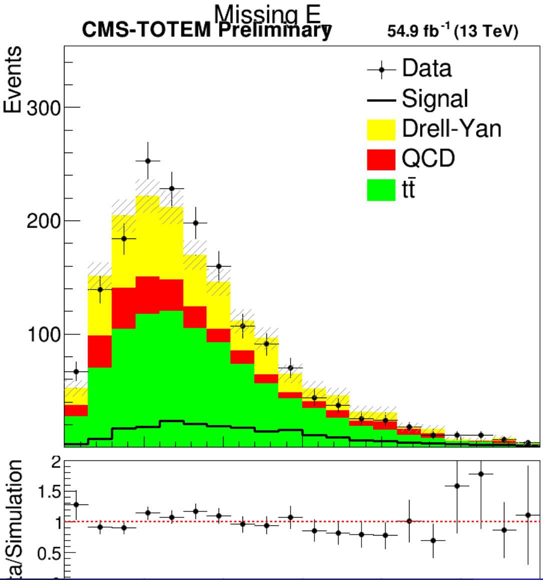
28. System Rapidity (after proton requirements)



29. Tau pT (after proton requirements)



30. MET (after proton requirements)



31. sinal.py: SFs Corrections

Muon Scale Factors Applied

SF Type	Function	Description
Trigger	`muon_trig_sf()`	IsoMu24 trigger efficiency
ID+Iso	`muon_idiso_sf()`	Identification and isolation
Reco	`mu_reco_sf()`	Reconstruction efficiency

Tables loaded from POGCorrections/

- `muon_neg18_lowTrig.txt` - Negative muon, pT $<$ 200 GeV
- `muon_pos18_lowTrig.txt` - Positive muon, pT $<$ 200 GeV
- `muon_18_highTrig.txt` - High pT muons ($>$ 200 GeV)
- `muon_IdISO.txt` - ID and isolation corrections

32. sinal.py: Event Selection cuts applied

Cut	Requirement
Basic	At least 1 muon and 1 tau
ID	`tau_id_full > 0.5` and `mu_id > 0.5`
Eta	`tau_eta < 2.4` and `mu_eta < 2.4`
Delta R	Angular separation > 0.4
pT	`tau_pt > 100 GeV`, `mu_pt > 35 GeV`
Charge	Opposite sign (product < 0)
Protons	Exactly 2 reconstructed protons

33. BDT summary: workflow

Pipeline

```
corrs_matrix.py --> TMVAClassification.C --> TMVAClassificationApplication.C
(feature QA)        (training)               (evaluation)
```

Files

File	Purpose
`corrs_matrix.py`	Correlation heatmaps from `variables2.txt` to select features
`TMVAClassification.C`	TMVA factory: train BDT + Likelihood + Fisher + MLP + PDERS
`TMVAClassificationApplication.C`	TMVA reader: apply trained weights, produce MVA score histograms
`dataset/weights/*.weights.xml`	Serialized trained models

34. BDT summary: training inputs and variables

Input samples

Role	File	Tree
Signal	MuTau_sinal_SM_2018_july.root	tree
Background	background_total-protons_syst.root	tree

Event weights in TMVA

```
dataloader->SetSignalWeightExpression("weight*weights_bsm_sf[51]");
dataloader->SetBackgroundWeightExpression("weight");
// Global scale factors:
double signalWeight = 0.00021;   double backgroundWeight = 104.;
```

9 training variables

Variable	Description
`sist_rap`	System rapidity
`acop`	System acoplanarity
`sist_pt`	System transverse momentum
`mu_pt`, `tau_pt`	Lepton transverse momenta
`sist_mass`	Invariant mass m(mu,tau)
`met_pt`	Missing transverse energy
`sist_mass - sqrt(13000^2 * xi_arm1_1 * xi_arm2_1)`	Mass matching (central vs proton)
`sist_rap - 0.5*log(xi_arm1_1/xi_arm2_1)`	Rapidity matching (central vs proton)

The last two compare the central mu+tau system with the forward proton kinematics. For true exclusive events they peak at zero.

35. BDT summary: configuration and output

BDT (AdaBoost) hyperparameters

`NTrees=850, MaxDepth=3, MinNodeSize=2.5%, AdaBoostBeta=0.5, UseBaggedBoost, BaggedSampleFraction=0.5, GiniIndex, nCuts=20`

Train/test split

- 6500 signal + 15000 background for training, rest for testing
- Random split, `NormMode=NumEvents`
- No additional preselection cuts inside TMVA

Application (TMVAClassificationApplication.C)

- Reads trained weights from `dataset/weights/`
- Computes derived matching variables in the event loop from `xi_arm1_1`, `xi_arm2_1`
- Per-event `weight` branch used as tree weight
- BDT score histogram: 20 bins in [-0.8, 0.8]
- Output: `TMVApp_sinal.root` (contains `MVA_BDT`, `MVA_Likelihood`, `MVA_Fisher`, `MVA_MLP`, `MVA_PDERS`)